EED-201 TUTORIAL 9 Answers

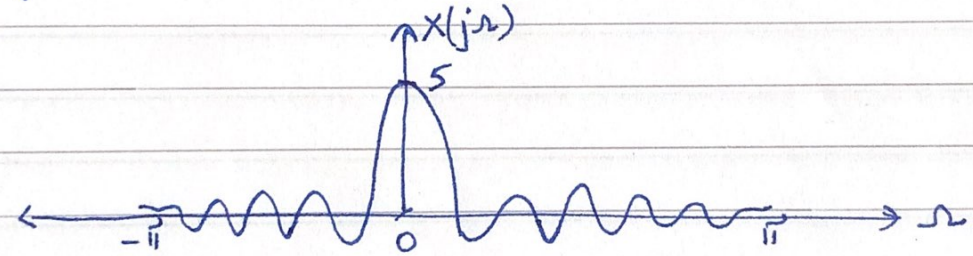
Ans-1

$$X(j\Omega) = \frac{\sin\left[\Omega(N_1 + 1/2)\right]}{\sin(\Omega/2)}$$

For $N_1 = 2$: $X(j\Omega) = \dfrac{\sin 2.5\Omega}{\sin 0.5\Omega}$

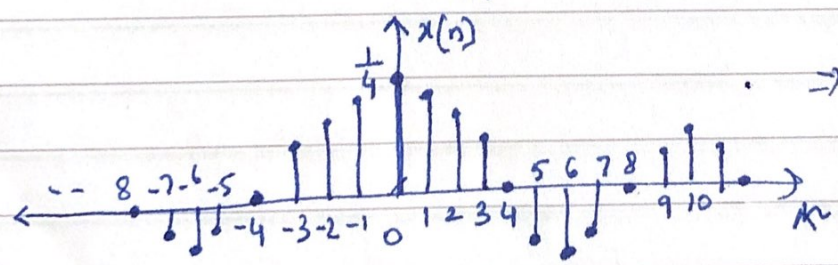
For $x(0.5n) \xleftrightarrow{DTFT} \underbrace{X(\Omega/0.5) = \dfrac{\sin 5\Omega}{\sin \Omega}}$

Plot of $X(\Omega/0.5)$ is shown below –



Ans-2

$$x[n] = \frac{\sin Wn}{\pi n}$$



$\Rightarrow$ for $W = \pi/4$

Ans-3 $x[n] = \text{I·DTFT of } X(\Omega) = \dfrac{1}{(1-ae^{-j\Omega})^2} \quad |a|<1$

$$x[n] = (n+1)\,a^n u[n]$$

Ans-4 $u[n] \xleftrightarrow{DTFT} \pi\delta(\Omega) + \dfrac{1}{1-e^{-j\Omega}} \quad ; \; |\Omega| \le \pi$

Ans-5 (i) $H(\Omega) = \dfrac{1+\frac{1}{2}e^{-j\Omega}}{1-\frac{1}{2}e^{-j\Omega}}$ (freq. response)

$$h[n] = \left(\tfrac{1}{2}\right)^n u[n] + \tfrac{1}{2}\left(\tfrac{1}{2}\right)^{n-1} u[n-1]$$

(ii) $y[n] = \cos\left[\pi/2\, n - 2\tan^{-1}(1/2)\right]$ $\quad \tan^{-1}(1/2)$

Ans-6 $\omega_c = 3\pi/4$

Ans-7 $x[n] = \begin{cases} 1, & n \le -2 \\ n+3, & -1 \le n \le 1 \\ 4, & n \ge 2 \end{cases}$